



Abhishek Kushwah
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B.Tech.
Gender: Male
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Examination	University	Institute	Year	CPI/%
Matriculation	State Board (M.P.)	I.P.S School	2018	97.40%
Intermediate	State Board (M.P.)	Govt. School Khareih	2020	90.40%

Scholastic achievements

- Secured **98.53** percentile in JEE Advanced and **99.52** percentile in JEE Mains among **0.93+** million aspirants (2021)
- Secured the **top(1st)** rank in entire region (**1000+** students) in the state Boards of **12th**-grade examination (2020)

Professional Experience

Data Scientist (Platform Team)

[July'25-Present]

L&T Finance Pvt. Ltd. | Streamlined cross-department automation workflows across the company.

- Worked on **24/7 automated logging** system sending pipeline error alerts with solutions, cutting **costs** by **50%**
- Developed a **chat-based Credit Risk & Collections Intelligence system** using **MCP** and **A2A** architectures, enabling **non-technical** users to query and analyse **10+** million rows data by simple natural-language conversation.
- Built an automated **email management system** using **Transformer embeddings & SVM** classification achieving **91%** accuracy, generating response, processing **1,500+** customer emails daily with automatic **assigning**
- Fine-tuned a **company-specific SLM** using **Q-LoRA** and **Reinforcement Learning** based instruction tuning for retrieval and better chatbot response, achieving **0.87 cosine similarity**, **0.39 BLEU**, and **0.51 ROUGE-I**

Internships and Projects Experience

AI (Finance Agentic Systems) & Data Intern (Vidur Research)

*Part time [Sep'25-Dec'25]

- Designed a **MAS**-based agentic retriever that queries **1M+** news data, **fundamentals**, and **100+ web pages**
- Fine-tuned **DeepSeek-VL-R1** model with **QLoRA** to precisely get numerical values from documents (**200+** words)
- Generated **Nifty50** investment reports in **<10 min** with **8.5/10** GPT-rated data **hygiene** using a **fine-tuned** model.

Generative AI(Freelance & Research)

[March'25–Nov'25]

- Freelance**-Built an agentic **AI orchestrator** integrating **25+** travel **APIs** (Flights, Hotels, Trains) using LLM-based task decomposition, implemented via LangGraph/LangChain and a **custom** architecture for speed and flexibility.
- Autonomous IDE**: Chat-driven system (**FastAPI + LangGraph**) with multi-agent workflows, generating **30+ .jsx** files per project, running multiple **<7B** models locally, and achieving **<50ms** streaming with **<800ms** live previews.
- Built a silent-speech recovery model using **LipNet + 3D CNN-GRUs**, delivering fast and accurate **real-time TTS**
- Implemented **MAV** detection in **C++**, detecting **UAVs** at **89% accuracy** and **25 FPS** using motion & appearance

Deep Learning Intern (Guide: Prof. Nirmal Punjabi)

[Jul'24-Sep'24]

- Built **LSTM**-based **time-series** model to forecast **6-month** demand trends; achieved **MAPE: 2.54%**, **RMSE: 10.54**.
- Developed NLP pipeline (**TF-IDF**, **LDA**, **PCA**, **KMeans**) to analyze data and customer sentiment; got **~20%** pricing gaps

Computer Vision Internship (Pharma Ilama Pvt. Ltd.)

[Jun'24-Aug'24]

- Manually created a dataset of **0.4 million** images with **labels** and **bounding boxes** to detect general medicines
- Used various models (Faster-RCNN, YOLOv8), got **Faster-RCNN** with the high accuracy (**95.5%+**) and **speed**

Research Projects

Revolutionizing Refueling: Automatic Fuel Filling Arm (Guide: Prof. Anirban Guha) [Aug'24-Nov'24]

- Developed techniques for detecting **fuel tanks** (**>85%** confidence) and determining their angles to guide the **robotic arm**
- Fine-tuned pretrained YOLOv8 for **100+** epochs, with **~0.6s** per-image inference time, cutting fuel filling time by **~30%**

High Accuracy Satellite Image classification (Guide: Prof. Biplab Banerjee)

[Apr'24-Aug'24]

- Increased accuracy by **(1.5-2%)** using **Drop-Block** with various **(4+)** CNN architectures in **classifying satellite** images
- Achieved remarkable closeness (**97%**) across multiple datasets, including **Pavia University**, **Houston**, **India Pines**,
- Used methods like **EfficientNet B7**, **Integrated ViT**, **Multihead attention**, enhanced **69.4%** for **Resnet-18** on CIFAR100

Junior Propulsion Engineer (Position of Responsibility- Hyperloop Tech Team)

[Aug'22-Feb'24]

- Contributed to propulsion team for IIT Bombay's first **hyperloop pod**, focusing on speed (**500 km/h+**) and efficiency

Technical Skills

Languages: Python, JavaScript, Java (Basics), C/C++ (Complete DSA), MATLAB, SQL, R, HTML, CSS, ReactJs

Libraries: Pytorch, TensorFlow, Keras, OpenCV, ScikitLearn, Seaborn, Numpy, Pandas, Nltk, Yolo, Roboflow

* **ML courses:** Introduction of remote sensing and Image processing*, Introduction to Machine Learning*, Deep Learning - Theory and Practice*, Statistical Machine Learning and Data Mining*, 3 Specialization Courses in Machine Learning (Coursera), Embedded Control & Robotics, Numerical Analysis, Linear Algebra, Differential Equation, Computer Programming and Utilization, Quantum Physics

Extracurricular: Mentored **120+** dancers with **6-member** choreographer's team in Institute's **flagship** Dance show(2x)